RISK REGISTER

Customer Churn Prediction System

Internship: Artificial Intelligence & Machine Learning

Organization: RabTech Academy

Project: Responsible Customer Churn Prediction

1. Purpose

This risk register identifies potential risks associated with the educational customer churn prediction prototype, assesses their qualitative likelihood and impact, and records proposed safeguards. Ratings are preliminary and should be reviewed by the responsible project or business team.

2. Risk assessment

ID	Risk	Likelihood	Impact	Priority
R01	Very small dataset (12 records) leads to unstable or misleading model estimates.	High	High	High
R02	Unclear data provenance, consent, or licence may make use or sharing unauthorized.	Medium	High	High
R03	Customer information may be exposed through insecure storage or sharing.	Medium	High	High
R04	Differences between subscription plans may lead to uneven errors or unfair treatment.	Medium	High	High
R05	False positive: a non-churned customer is incorrectly flagged as likely to churn.	Medium	Medium	Medium
R06	False negative: a churned customer is not identified by the model.	Medium	High	High
R07	Data leakage occurs if a feature contains information recorded after the churn outcome.	Medium	High	High
R08	Incorrect, outdated, or invalid feature values produce misleading predictions.	Medium	Medium	Medium

R09	Model output is used for automated or unfair customer decisions without human review.	Medium	High	High
R10	Customer behaviour or data patterns change, reducing model usefulness over time.	Medium	Medium	Medium

3. Mitigation and safeguards

R01 – Small dataset: Treat results as a learning demonstration only. Obtain a larger, representative dataset and perform suitable validation before deployment.

R02 – Provenance and permission: Ask the dataset provider to confirm source, collection method, permitted use, consent status, licence, and redistribution conditions. Do not publish the supplied CSV unless permission is clear.

R03 – Privacy and security: Limit access, use secure storage, avoid sharing identifiable information, and remove unnecessary identifiers from modelling inputs.

R04 – Potential bias: Inspect errors across plan types only when adequate data is available. Do not claim fairness from this 12-record sample; collect representative data and conduct a meaningful assessment.

R05 – False positives: Require human review before retention contact. Avoid unnecessary or intrusive messages, discounts, or campaigns.

R06 – False negatives: Review missed churn cases, assess the business cost of missed predictions, and improve data and model validation.

R07 – Data leakage: Confirm that every input feature is available at the prediction time. Exclude post-outcome information and keep preprocessing within the cross-validation pipeline.

R08 – Data quality: Validate ranges, categories, missingness, duplicates, and freshness before using data. Investigate unusual values.

R09 – Misuse: Use predictions only as decision support. Do not automatically cancel accounts, deny services, or change prices based on model output. Keep a human reviewer involved.

R10 – Model drift: Monitor data and errors over time. Reassess the model when customer behaviour or business processes change.

4. Proposed ownership and monitoring

Area	Proposed owner	Monitoring action
Data quality	Data analyst	Check schema, ranges, missing values, duplicates, and freshness.
Model performance	ML engineer	Review confusion matrix, precision, recall, F1-score, and errors.
Privacy and access	Data/security owner	Review permissions, access controls, and data-sharing practices.
Fairness	ML engineer / reviewer	Examine group-level errors only when sample sizes support analysis.
Business use	Business manager	Review proposed retention actions and prevent automated adverse decisions.

5. Rollback and contingency plan

1. Pause use of model predictions if data quality, privacy, fairness, or performance concerns are identified.
2. Return to manual or existing rule-based customer review while the issue is investigated.
3. Identify and document the cause, including affected data, features, model version, and decisions.
4. Correct the issue and repeat appropriate validation before resuming model-assisted review.
5. Keep a record of the review and the decision to resume or retire the model.

6. Limitations

Likelihood and impact ratings are qualitative preliminary judgments, not measured incident probabilities. The supplied dataset contains only 12 records, so model reliability and group-level fairness cannot be established. The original data provenance, consent, and licensing conditions were not documented in the supplied materials and require confirmation from the provider.

7. Conclusion

The principal risks are the very small sample, uncertain data provenance and permissions, potential privacy exposure, prediction errors, possible unfairness, and misuse of model output. The prototype should remain educational and should not be used for real customer decisions without additional data, documented permissions, appropriate validation, monitoring, and human oversight.